



MADRID AIR QUALITY

Group: 404 Sleep.exe

Data Loading and Inspection

Hourly observations from 24 stations across Madrid, 2001 to 2024

The stable-sensor comparison suggests that the NO₂ trend is not mainly affected by changes in the number of active monitoring stations.

64M

Total rows

24

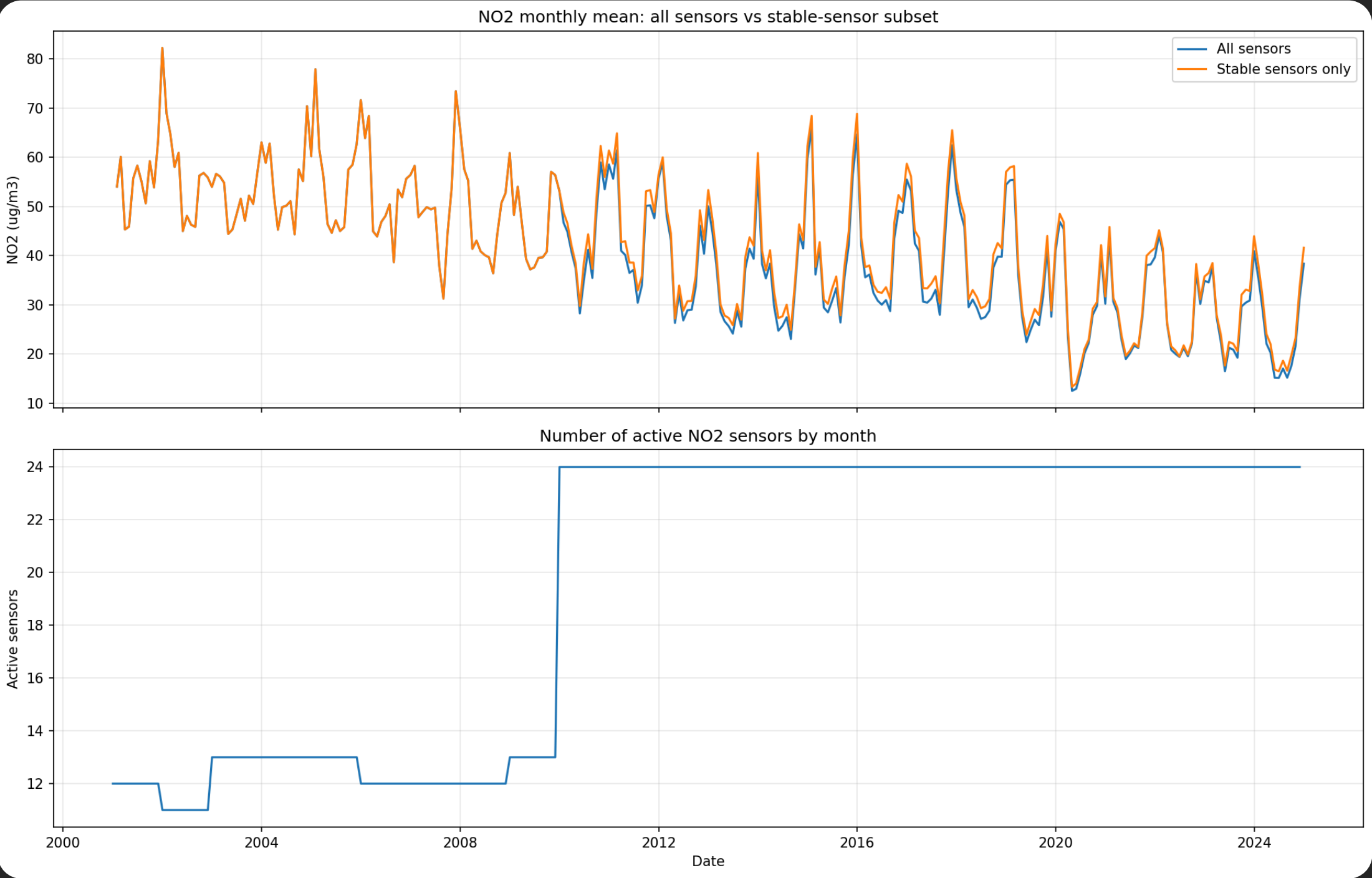
Active sensors

24

Years covered

30+

Variables tracked

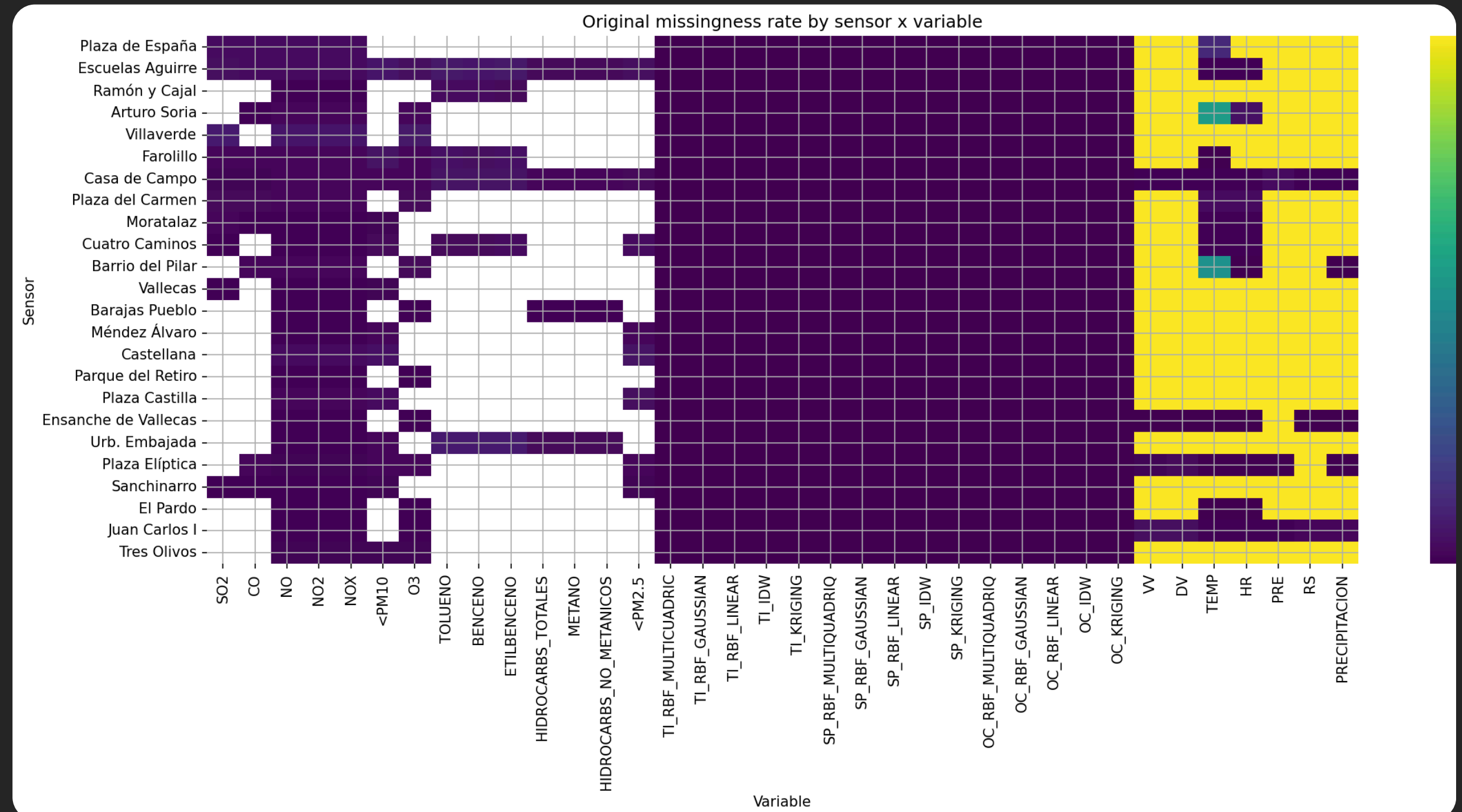


Missingness and Data Quality

Reconstructed gaps via the is_interpolated flag

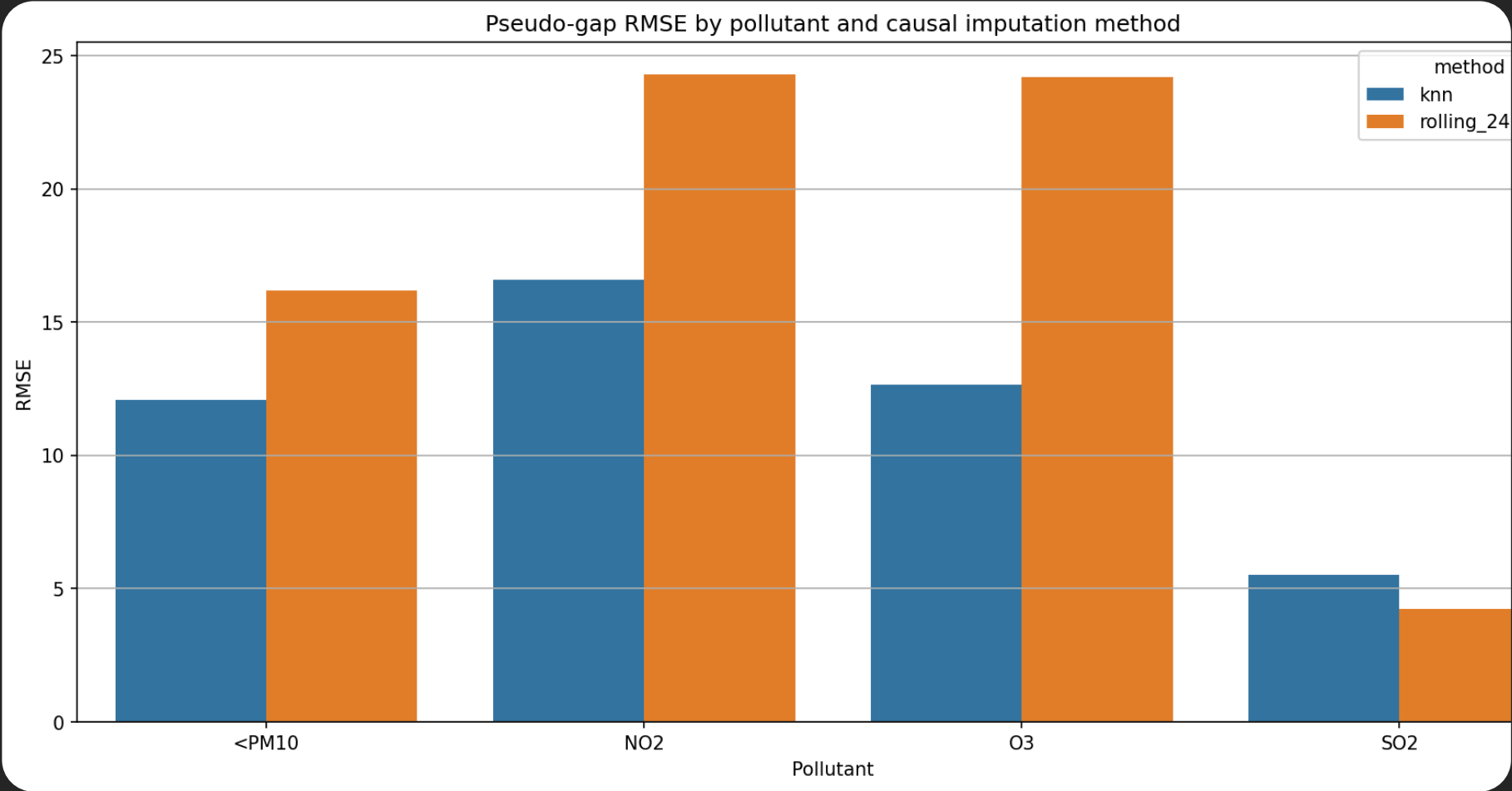
Our Findings

- **Air pollutants** are largely complete, gaps are sensor and period specific.
- **Meteorological variables** have the highest missingness due to limited station and time coverage.
- **Traffic** is mostly available across all stations.
- **Quality checks** flagged invalid HR (outside 0 to 100), DV (outside 0 to 360), negative non-negative values, IQR outliers and duplicate rows.

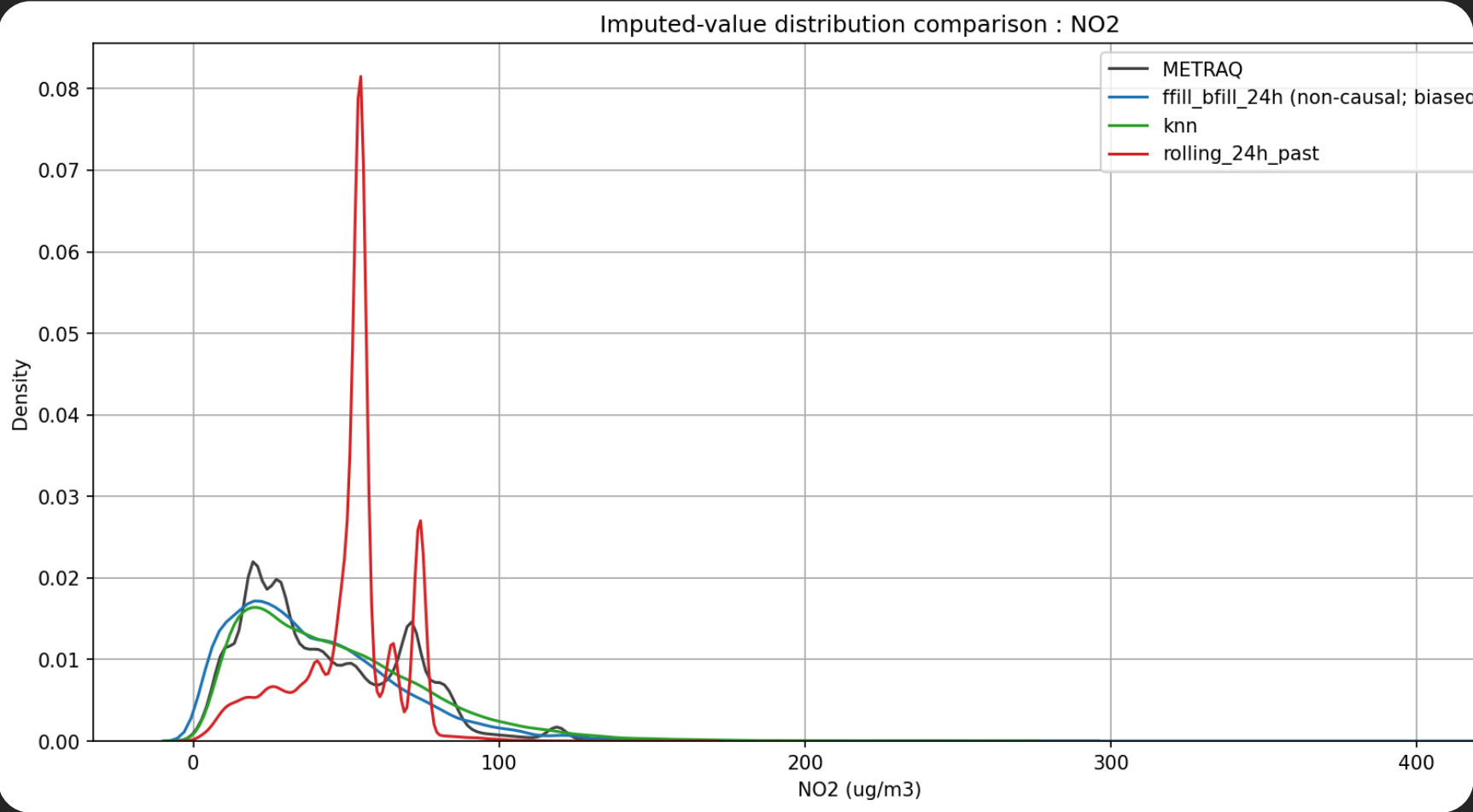


Imputation

Two causal methods compared against the METRAQ baseline



Pseudo-gap RMSE by method



Distribution of imputed values, NO2

N O 2

Best method: KNN (IDW)

O 3

Best method: KNN (IDW)

P M 1 0

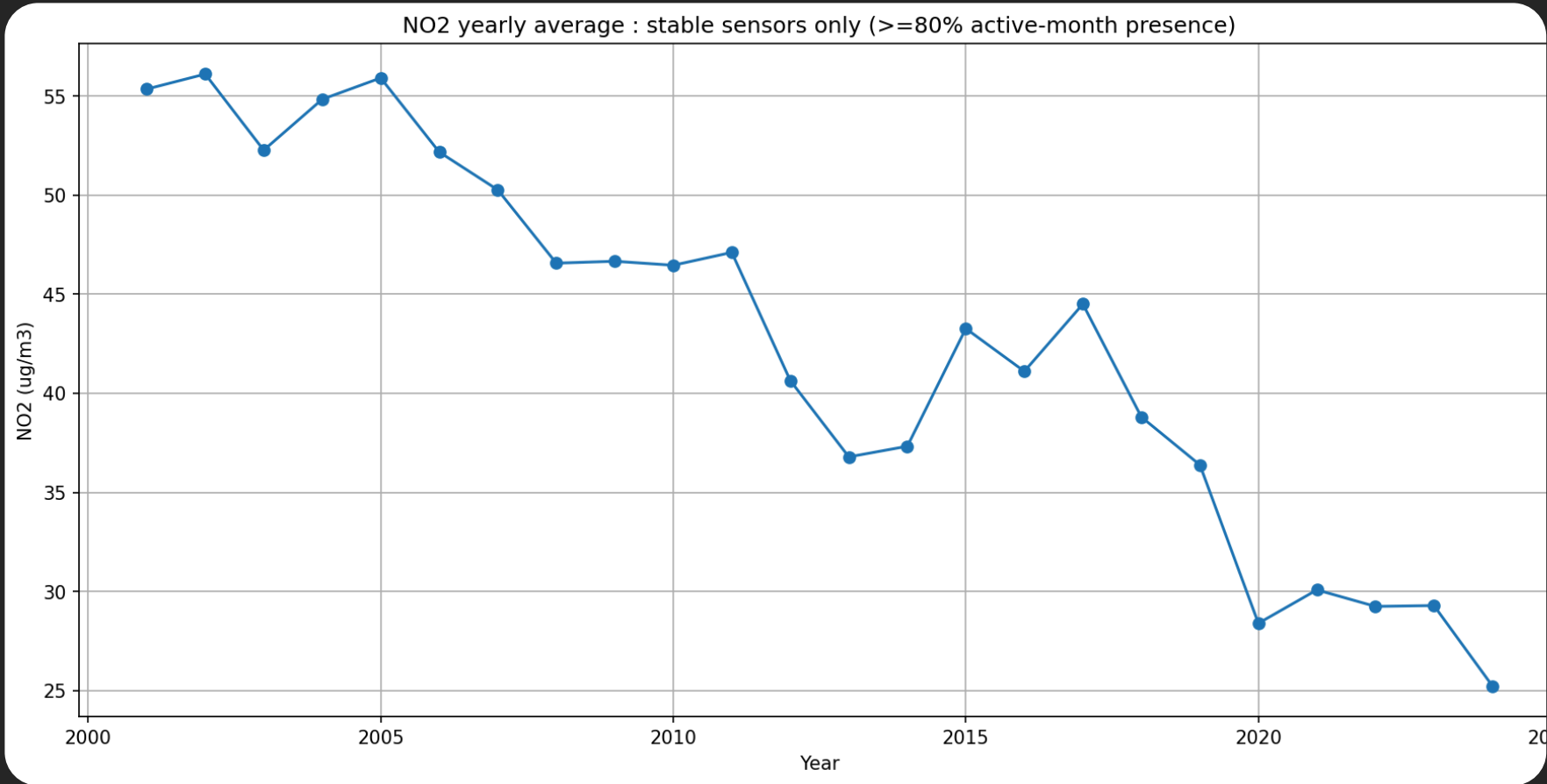
Best method: KNN (IDW)

S O 2

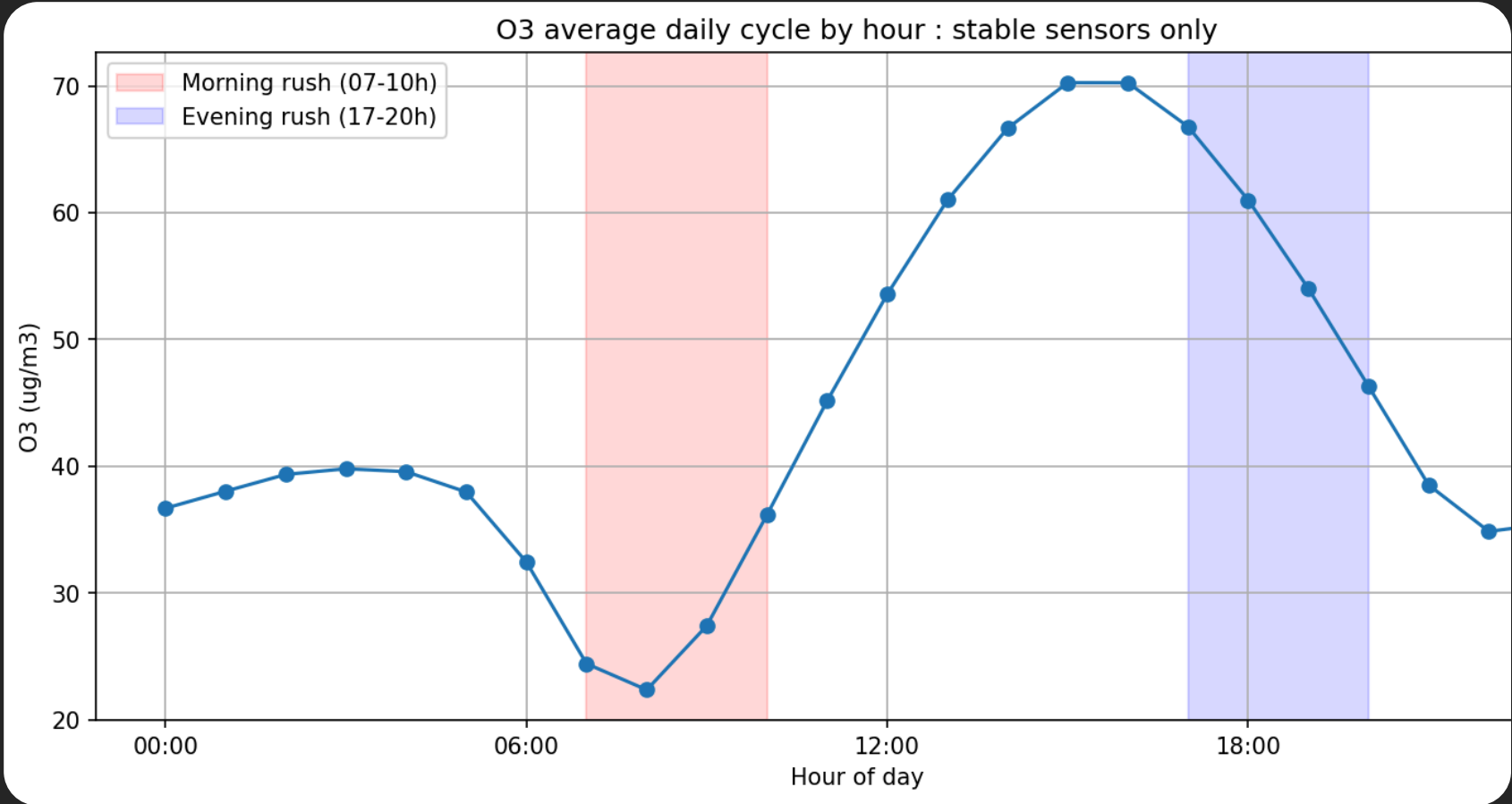
Best method: Rolling 24h

Temporal Analysis

Yearly trend, seasonality and diurnal cycle on stable sensors



NO2 yearly average, stable sensors



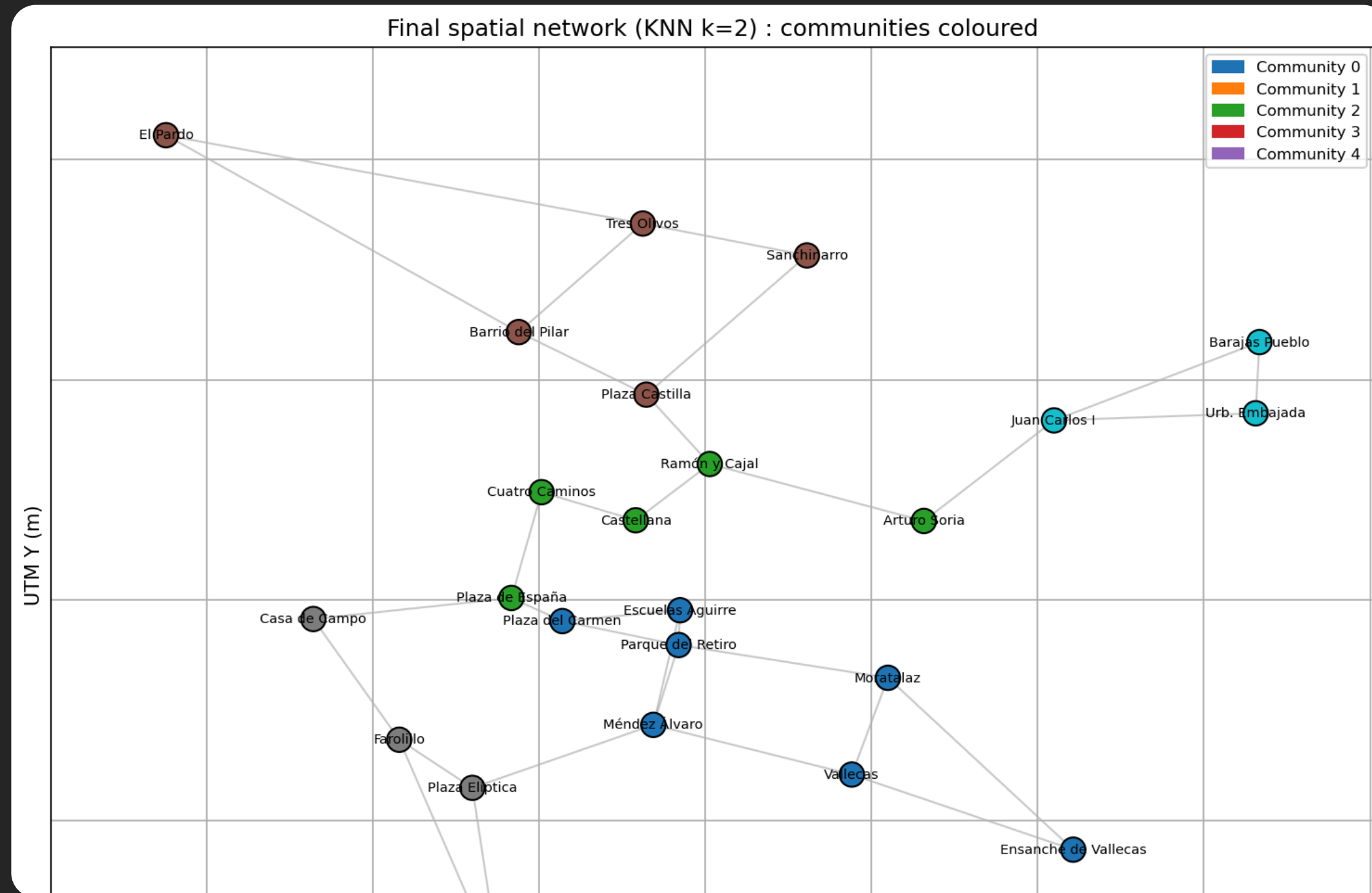
O3 average daily cycle

KEY FINDINGS

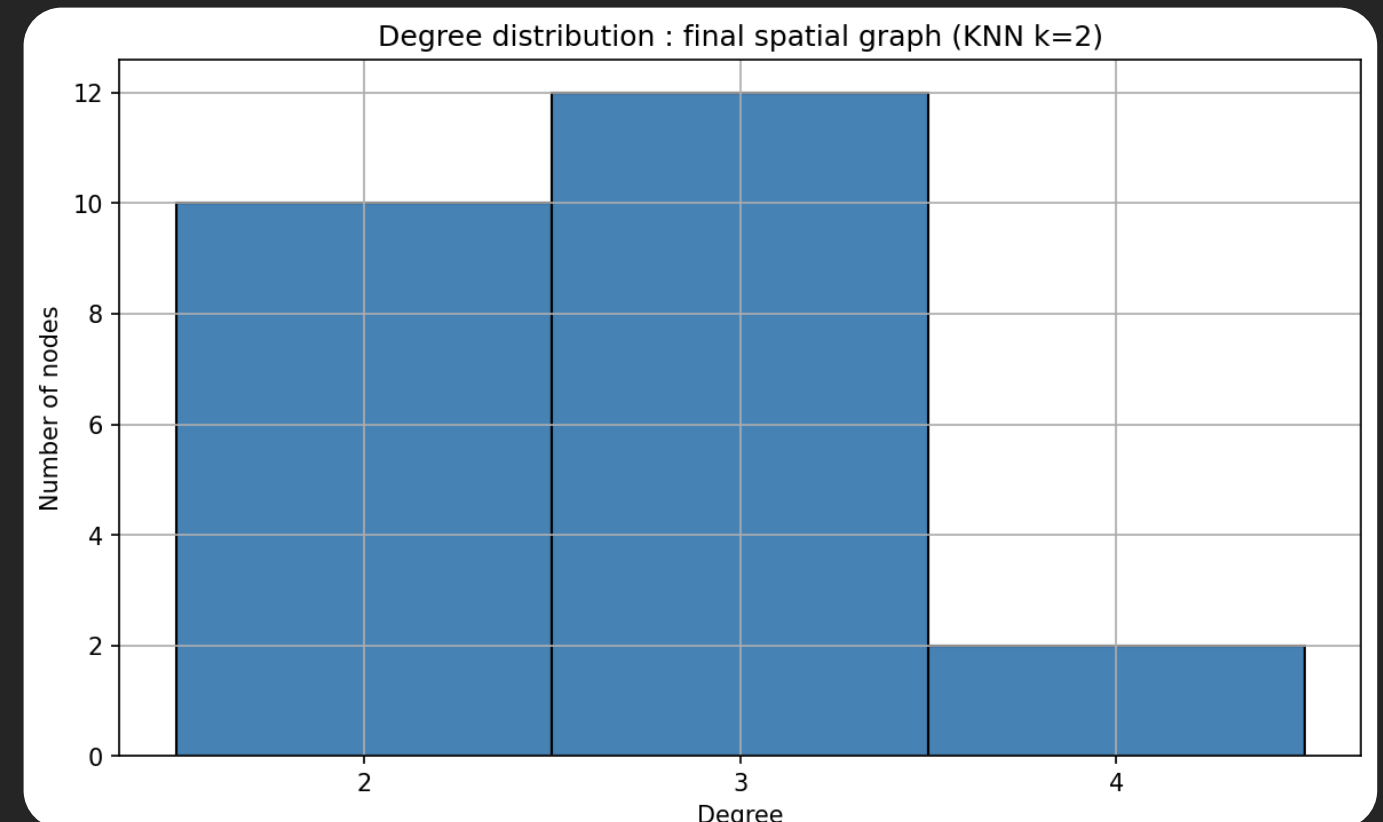
NO₂ shows a clear long-term decrease on the stable-sensor subset, indicating that the trend is not mainly driven by changes in station coverage. O₃ follows a strong daylight-driven cycle, peaking in the afternoon due to photochemical formation. This highlights that different pollutants follow different temporal patterns.

Spatial Network

KNN with $k = 2$ smallest k that yields a single connected component



Final graph: 24 nodes, $k = 2$, 5 communities matching geographic clusters (city centre, north, east, south, periphery).

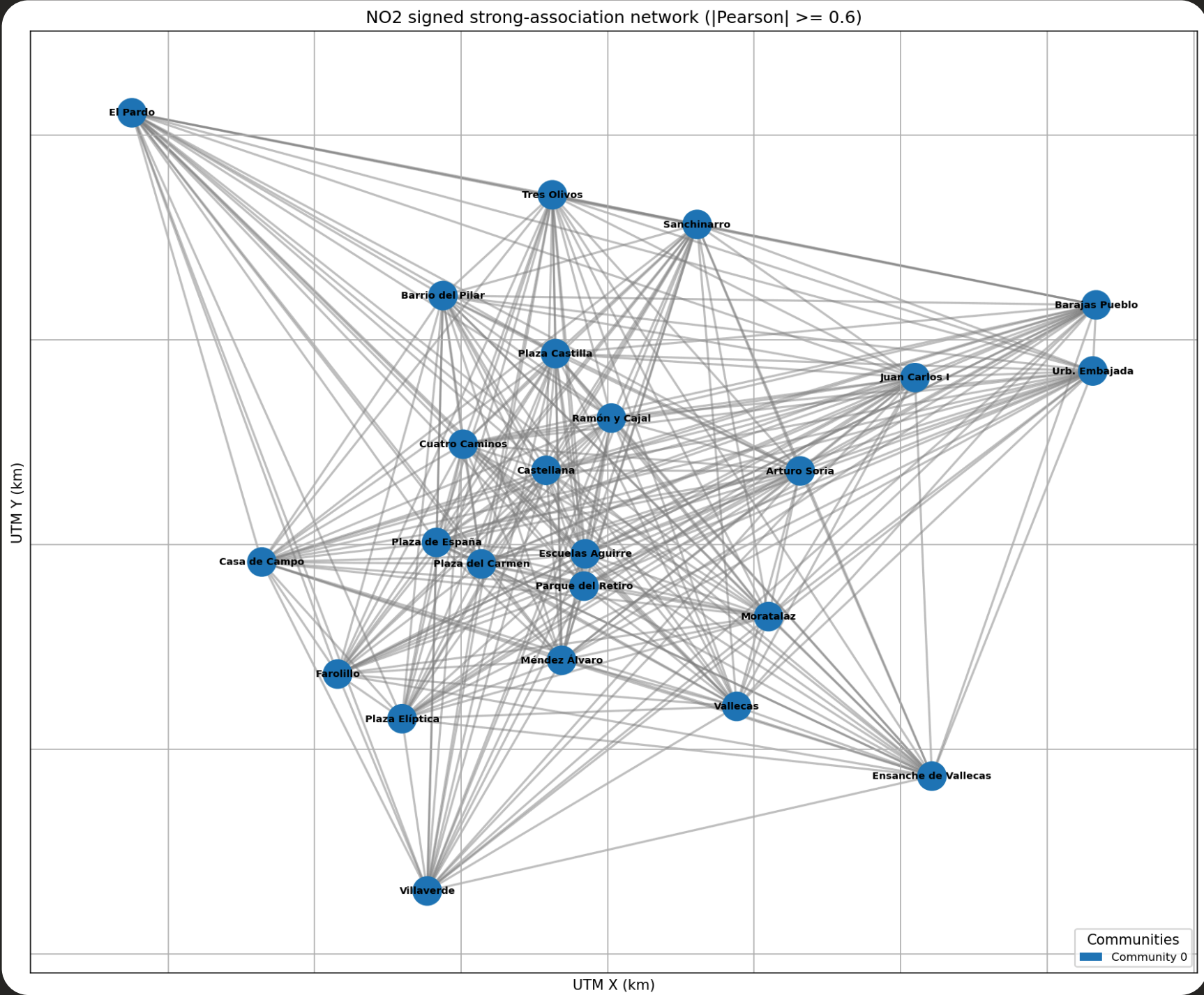
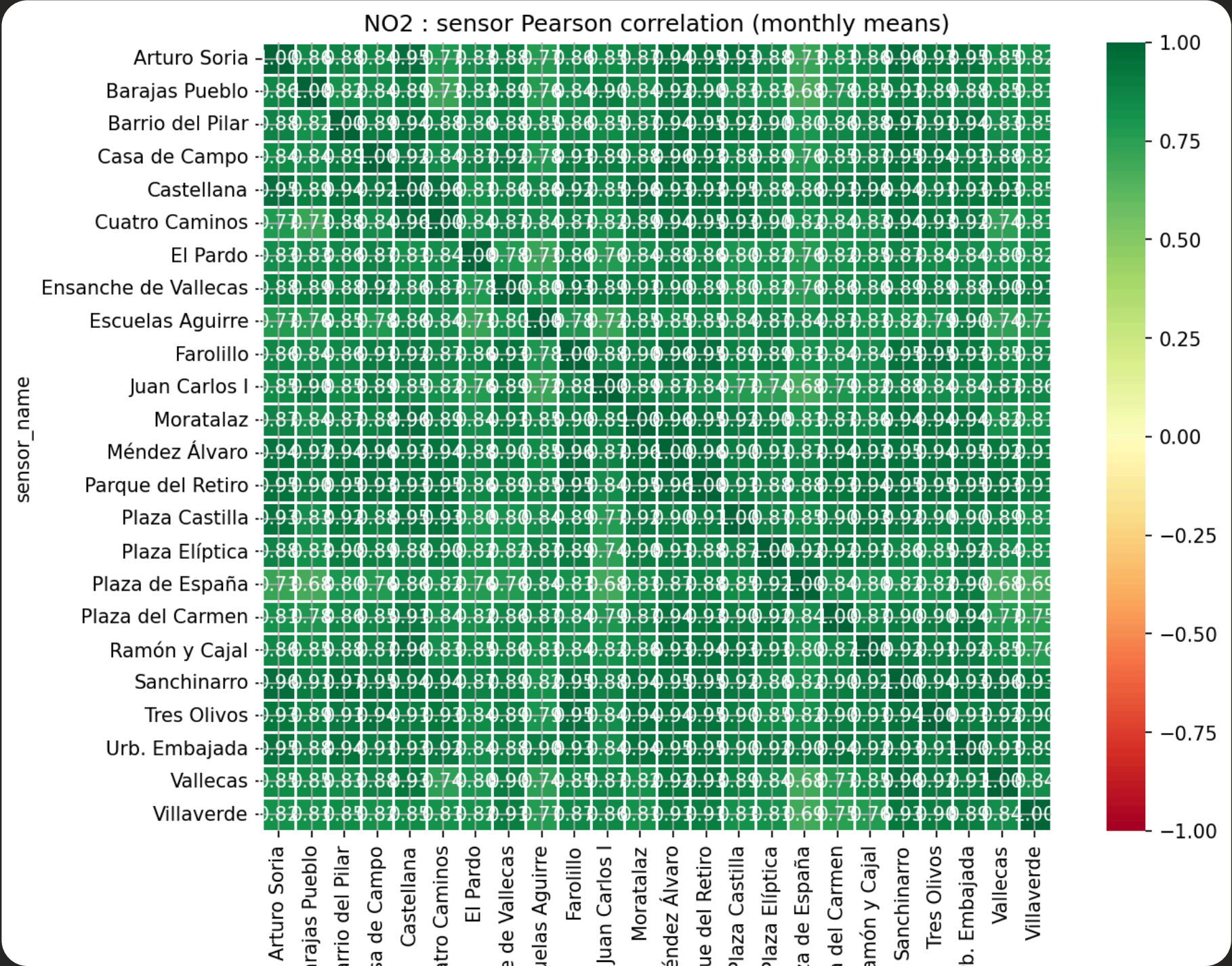


WHY KNN?

A simple distance threshold can make the network too dense because all stations are located within the same city. Using KNN with $k = 2$ preserves local neighbourhood relationships while still keeping the graph connected. This makes the network easier to interpret and allows meaningful comparison of degree distribution and community structure.

Correlation Network

Edges based on monthly Pearson correlation, threshold 0.6. The NO₂ correlation network is much denser than the spatial KNN graph. This means many stations follow similar monthly pollution patterns, likely due to shared traffic rhythms, seasonal cycles, and city-wide meteorological conditions.



Why Use a Graph-Based Diffusion Model?

PROPAGATION RULE

$$X(t+1) = (1 - \alpha) \cdot X(t) + \alpha \cdot W \cdot X(t)$$

1

NEIGHBOUR-BASED DIFFUSION

Discrete heat equation on the sensor graph. Pollutants do not stay still, they spread between neighbouring areas via wind and shared boundary conditions.

2

NORMALIZED WEIGHTS

Neighbour weights are normalized to sum to 1, so the graph step does not artificially increase predicted pollution values.

3

ONE TUNABLE PARAMETER

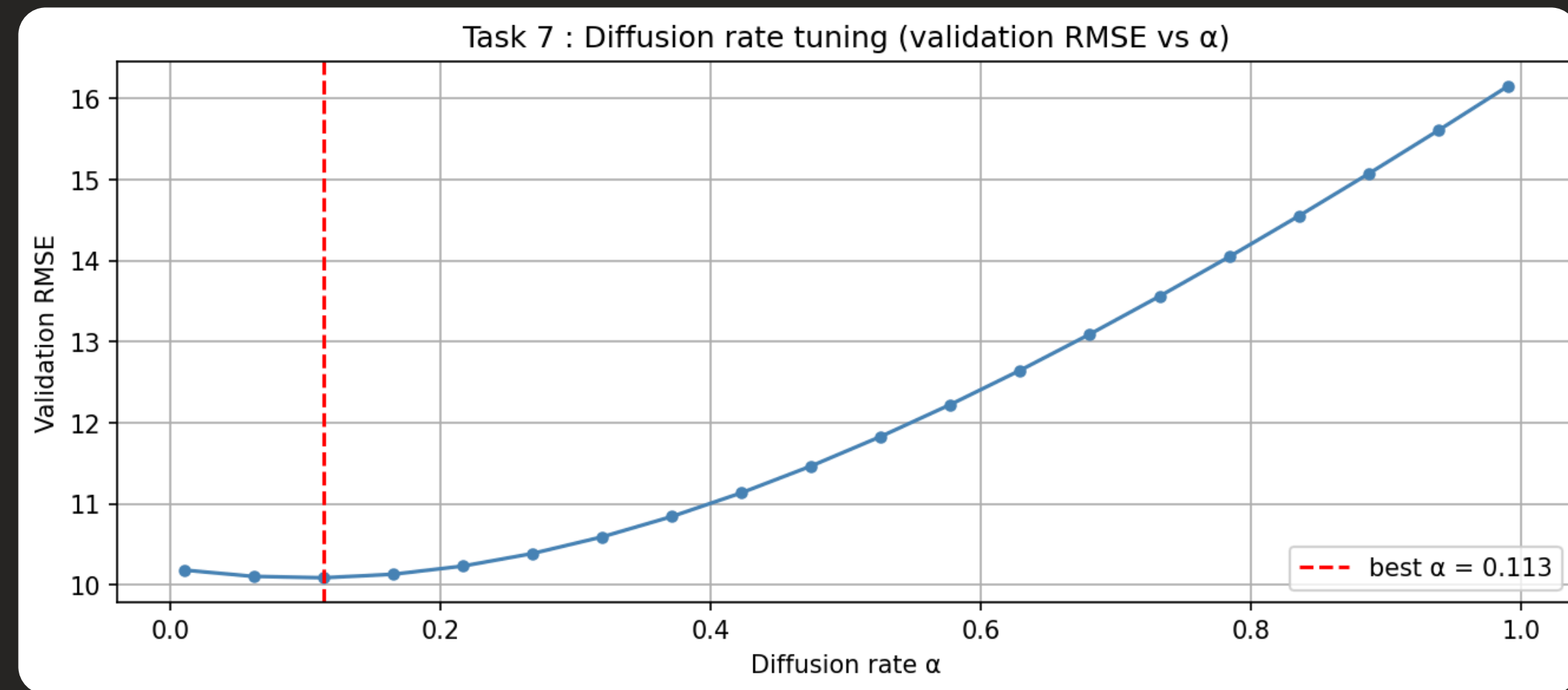
Only $\alpha \in (0,1)$ controls diffusion speed. Easy to interpret, easy to validate, hard to overfit on a small graph.

$\alpha \rightarrow 0$: sensors evolve independently (no spread)

$\alpha \rightarrow 1$: sensors fully average to their neighbours (instant mixing)

How we used the diffusion model

Graph weights are based on inverse distance, while α is selected through chronological validation.



1 Build W

We constructed a row-normalized spatial weight matrix using inverse-distance weights from the KNN graph.

2 Split data

The NO₂ time series was split chronologically into training, validation, and test periods to avoid temporal leakage.

3 Tune α

The diffusion rate α was selected through a grid search from 0.01 to 0.99, using validation RMSE as the selection criterion.

4 Evaluate

The selected model was evaluated on one-step and recursive forecasts, and compared against persistence and SAR Ridge baselines.

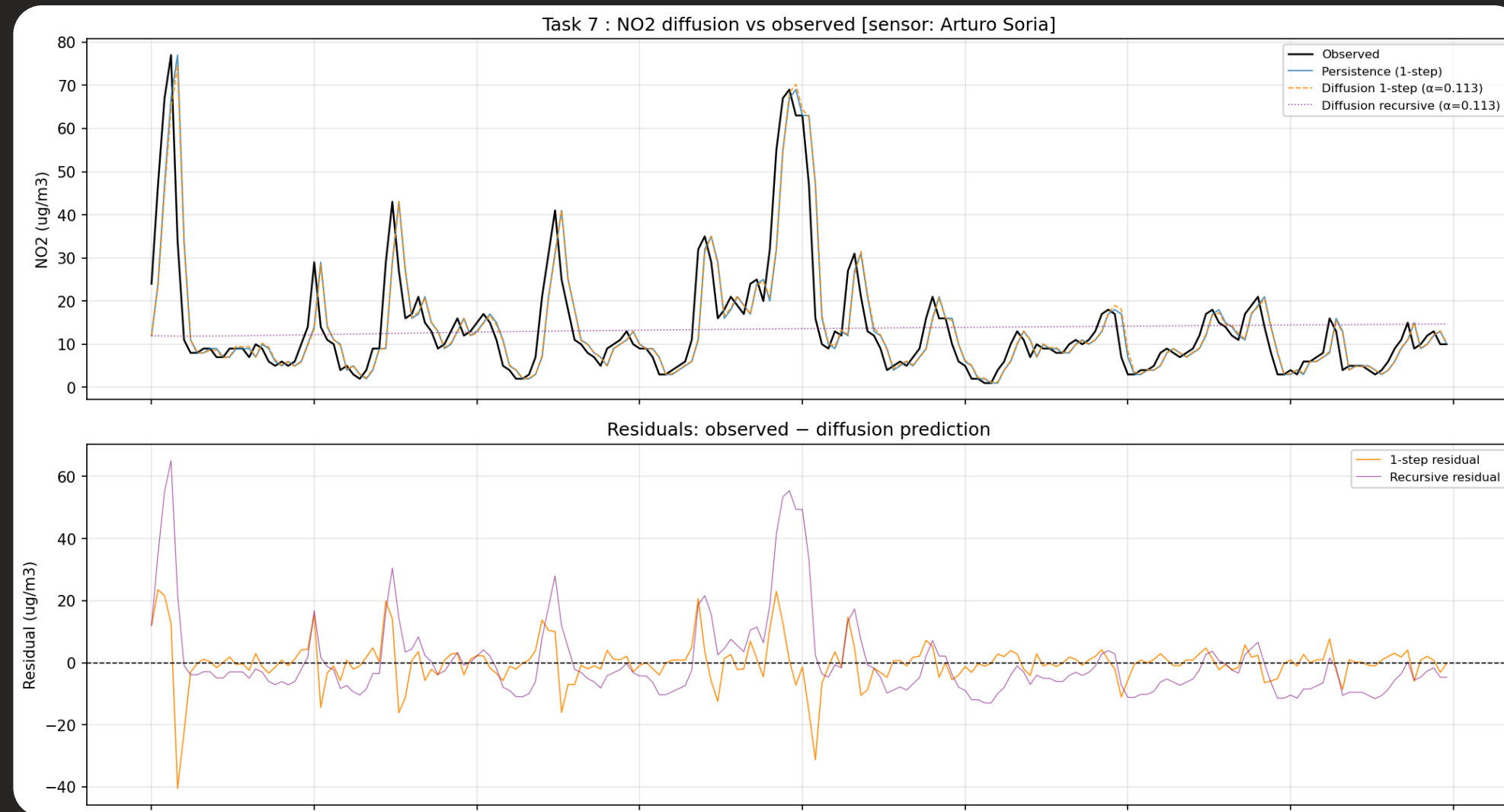
BEST DIFFUSION RATE

$\alpha = 0.113$

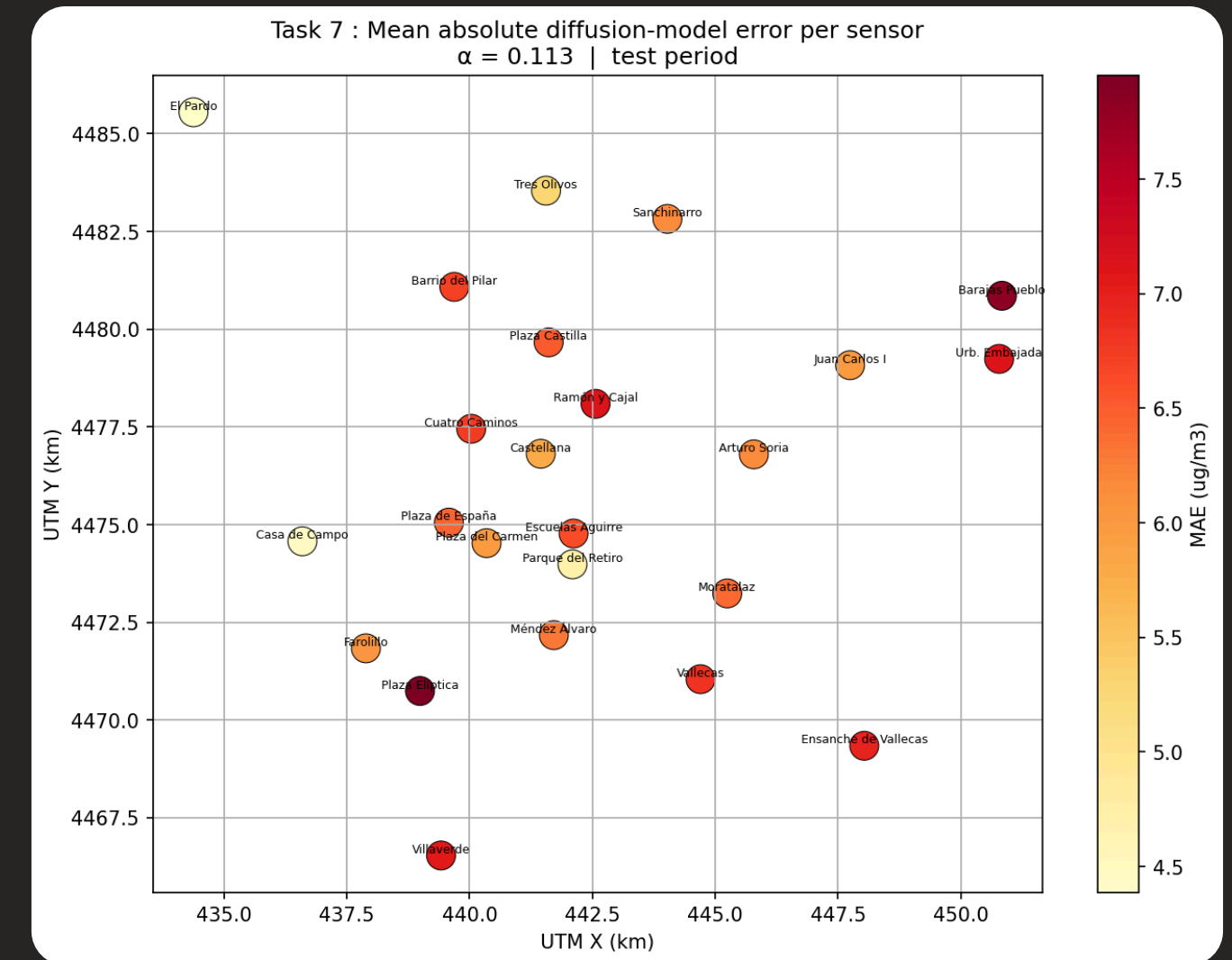
Each hour, approx. 11% of every sensor's predicted value comes from its graph neighbours, and approx. 89% from its own previous value. The selected α indicates that each hourly prediction relies mainly on the sensor's own previous NO₂ level, with a smaller contribution from neighbouring stations.

Diffusion model vs observed NO2

1-step diffusion tracks observations closely, recursive simulation degrades over time



NO2 diffusion vs observed (sensor: Arturo Soria) and residuals

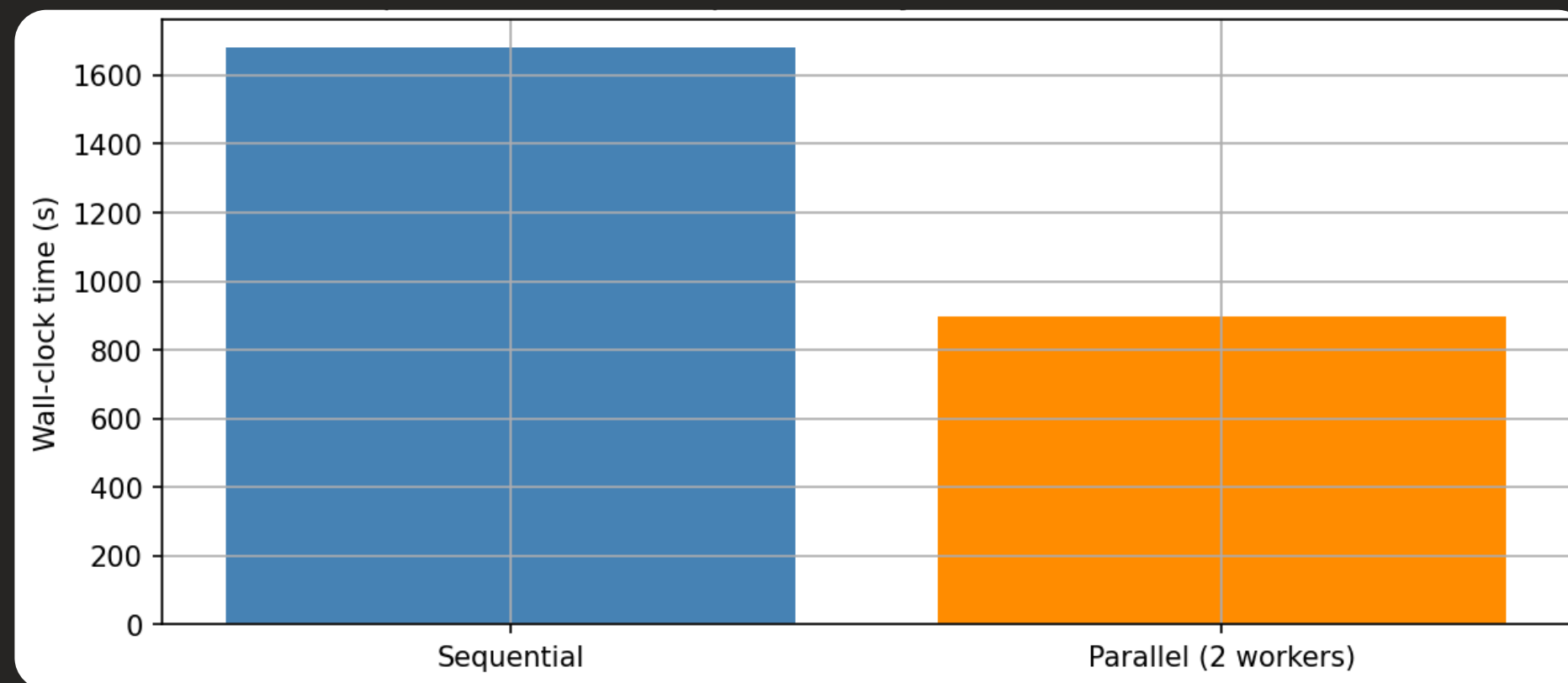


Mean absolute error per sensor

TAKEAWAY: The one-step diffusion model provides a reasonable approximation of observed NO₂ dynamics, but recursive predictions accumulate error over time. Higher MAE values at stations such as Plaza Elíptica, Barajas Pueblo, and Villaverde indicate that local station-specific factors are not fully explained by spatial diffusion alone.

Parallelization

Hourly correlation matrices computed for each (year, sensor) job



Parallelization reduced runtime by almost half. The remaining bottleneck is disk I/O from loading large yearly CSV files, using Parquet year partitions would improve scalability.

MEASURED SPEEDUP

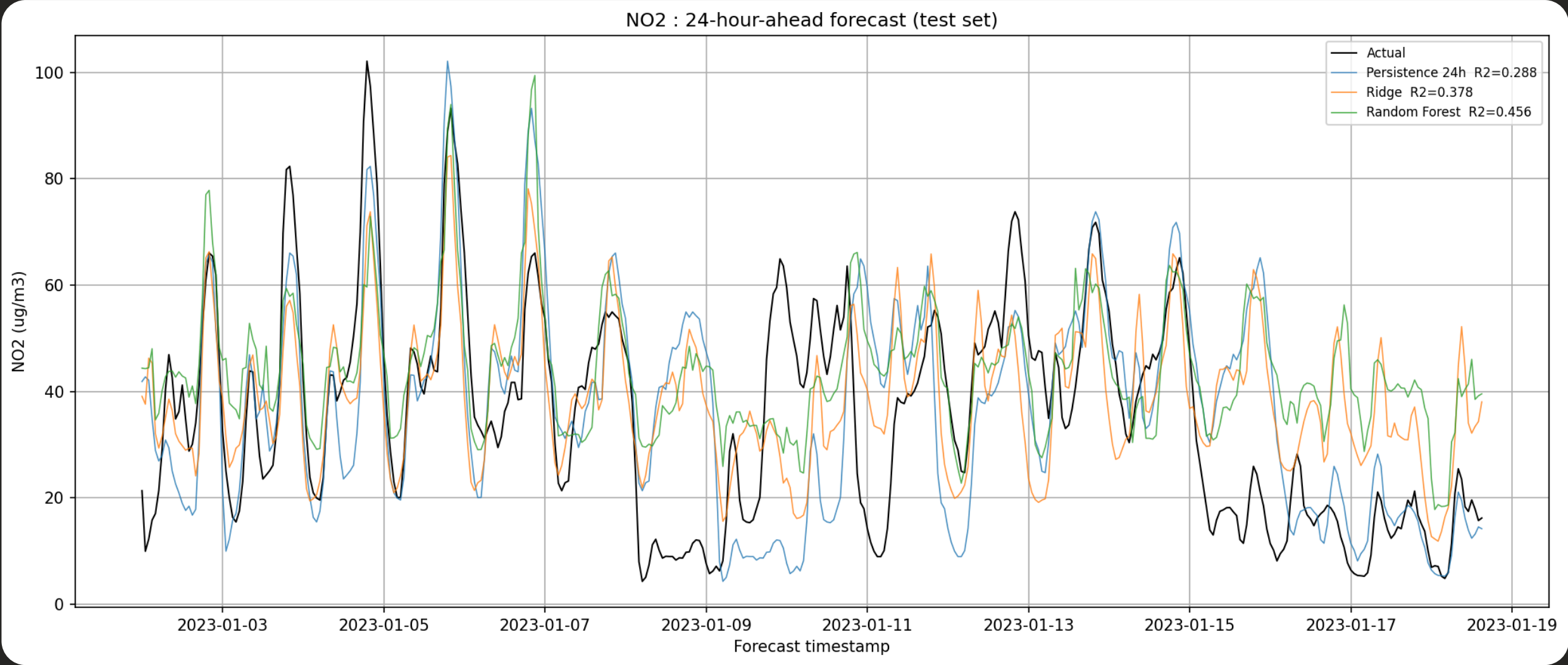
~ 1.9x with 2 workers

APPROACH

- Year-partitioned files, each worker reads only its slice.
- ProcessPoolExecutor runs independent year × sensor jobs.
- Chunked CSV reading keeps RAM manageable.
- Sequential and parallel outputs were compared using matrix hashes. Matching hashes confirm that parallelization improved speed without changing the correlation results.

24-hour-ahead Forecasting

City-level NO2 with lagged meteorology, traffic and calendar features

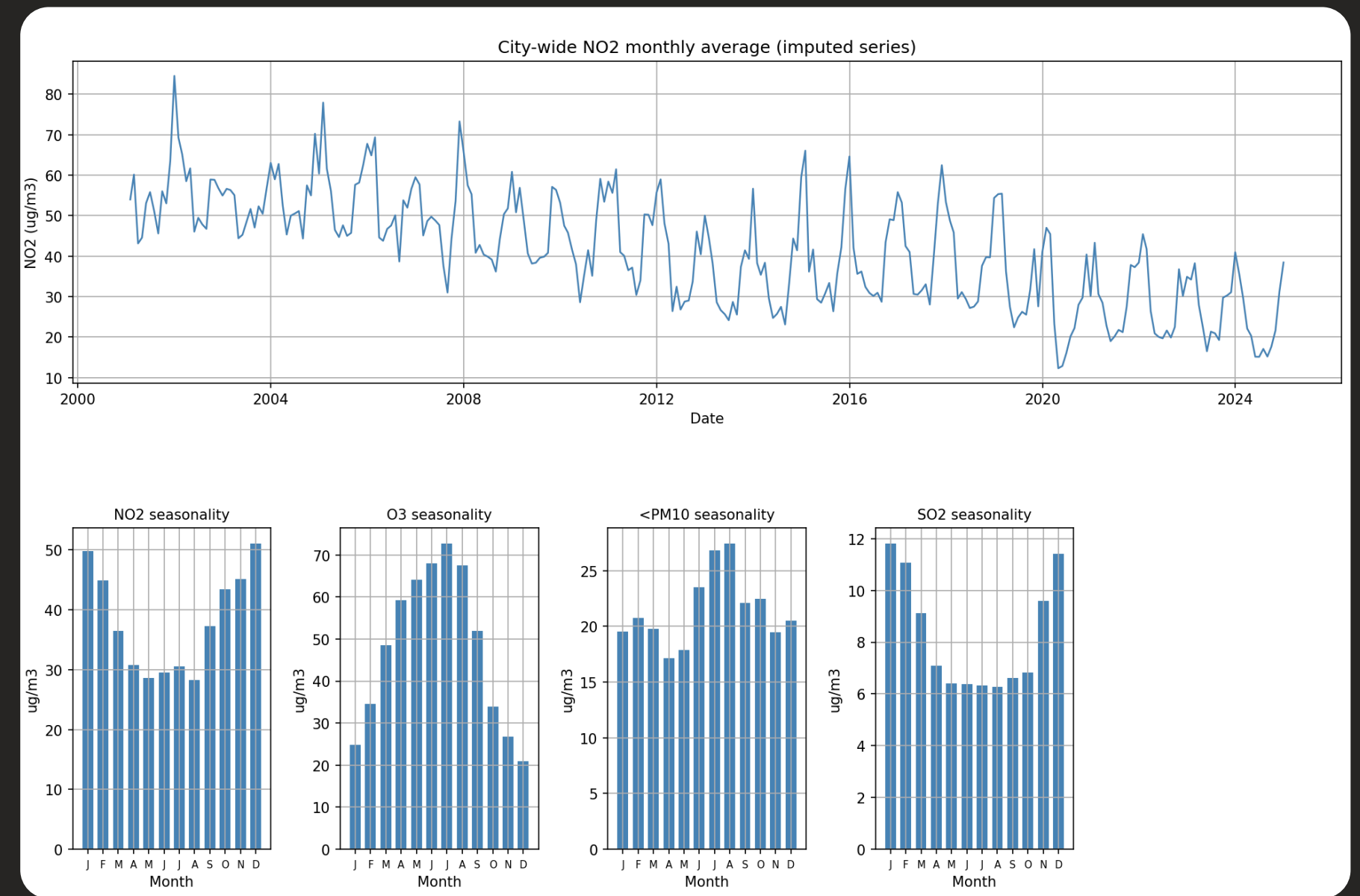
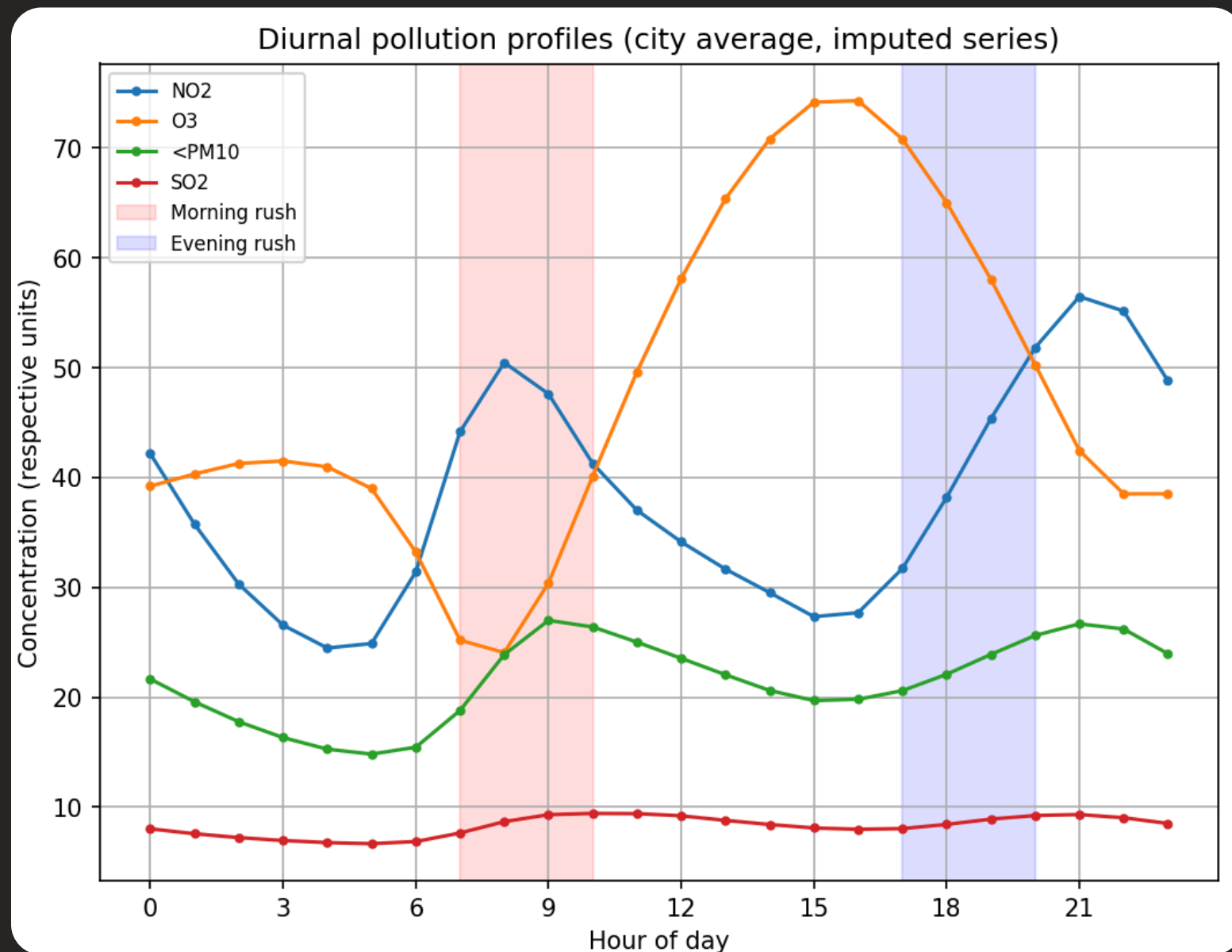


NO2 24h ahead forecast on the test set

Random Forest R^2	0.458
Ridge R^2	0.378
Persistence 24h R^2	0.288

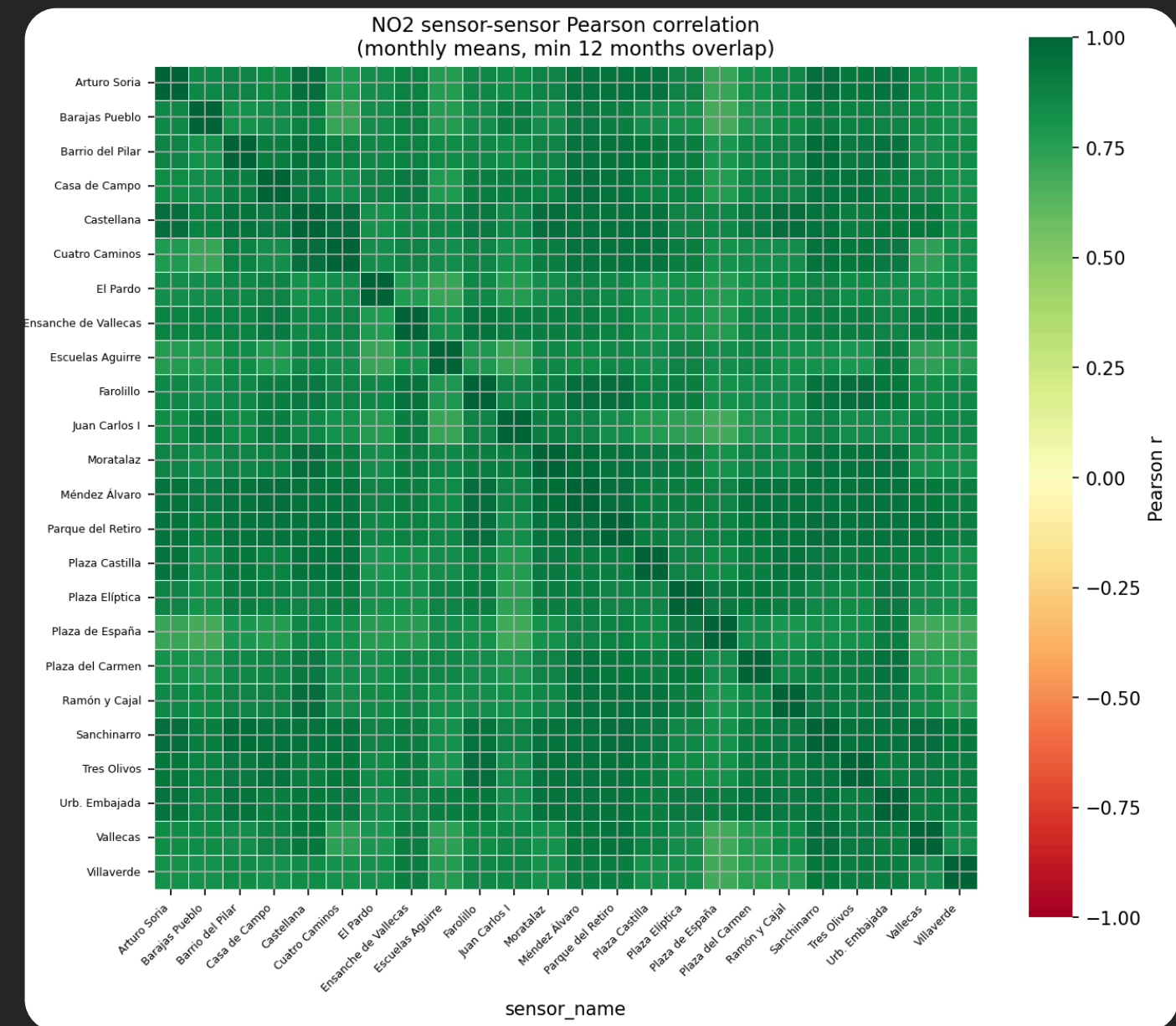
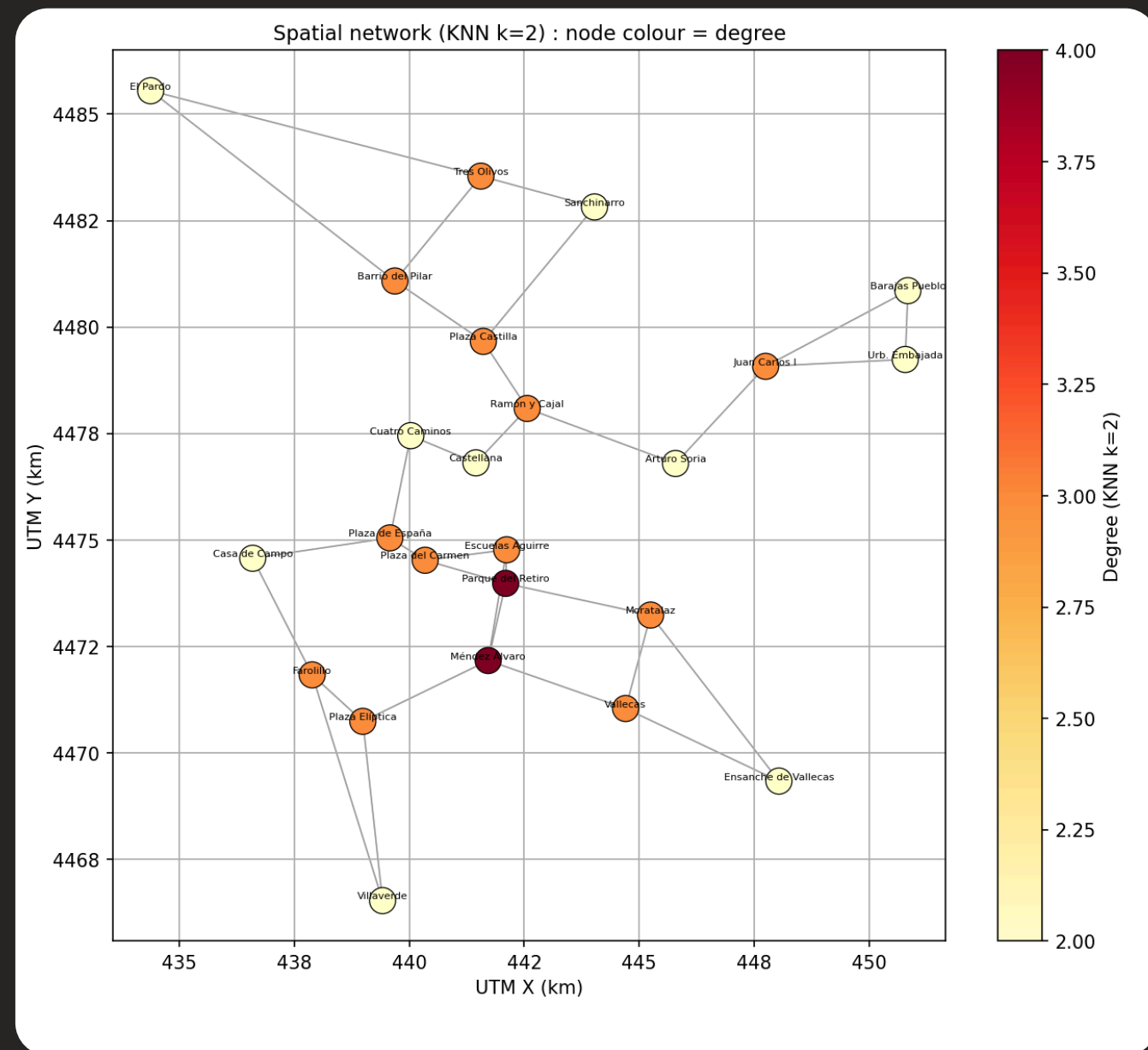
Random Forest shows the strongest forecasting performance, outperforming both Ridge regression and the 24-hour persistence baseline. The results indicate that lagged NO₂ has high predictive importance, suggesting strong temporal dependence in city-level pollution patterns. The improvement over Ridge also suggests that some relationships between predictors and NO₂ are non-linear.

Final Visualization: Pollution Patterns



The left panel shows daily pollution behaviour, NO_2 rises during traffic-heavy hours, while O_3 peaks later in the day due to sunlight-driven formation. The right panel highlights that meteorological variables have the highest missingness because they are available mainly in later years and at fewer stations.

Final Visualization: Spatial and Correlation Structure



The spatial graph shows how stations are connected by geographic proximity, while the correlation matrix shows whether their NO₂ patterns move together over time. The dense positive correlations suggest that Madrid's stations are influenced both by local distance and by shared city-wide factors.



THANK YOU

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